

How to Set up a Minikube Lab with VMware Workstation

Recently, I started looking at Kubernetes. As a first step, I went through the tutorial on how to [install Minikube](#). On this link, we get the instructions on how to set up Minikube on Windows 10 using either Hyper-V or VirtualBox as a hypervisor. However, my preference is to use VMware Workstation as the hypervisor. When I went through their tutorial, I found that the documentation for setting this up was not available on the [MiniKube vmware driver page](#) at the time of writing. After troubleshooting, I installed Minikube and managed to get it working with VMware Workstation. Here are the steps that I followed.

Install a Hypervisor

- Install [VMware Workstation 15.5](#).
- After installing VMware Workstation, we need to have **vmrun.exe** on your Windows PATH. Add **C:\Program Files (x86)\VMware\VMware Workstation** to your Windows PATH variable.
- As an example of how to do this, I referred to the link at [Add a folder to your Windows PATH variable](#).

Install Docker machine driver

- Download Docker machine driver **docker-machine-driver-vmware_windows_amd64.exe** from <https://github.com/machine-drivers/docker-machine-driver-vmware/releases>.
- Create a folder at **C:\Program Files\docker-machine-driver**.
- Copy **docker-machine-driver-vmware_windows_amd64.exe** to this folder.
- Rename **docker-machine-driver-vmware_windows_amd64.exe** to **docker-machine-driver-vmware.exe**.
- Add this folder to your Windows PATH variable as above.

Install kubectl

- These steps are distilled from the instructions at [Install and Set Up kubectl](#).
- Download the [latest version of kubectl](#) for Windows/AMD64.
- Create a folder at **C:\Program Files\Kubectl**.
- Add the **kubectl.exe** to this folder.
- Add this folder to your Windows PATH variable.

Check the version

```
kubectl version --client
```

Install Minikube using an installer executable

These instructions are distilled from the MiniKube documentation. Please note that I specify the hypervisor driver as **vmware**. To install Minikube manually on Windows using the Windows Installer for AMD64, [download minikube-installer](#) and execute the installer.

Start the Minikube

Use vmware as the hypervisor driver

```
minikube start --driver=vmware
```

Example output

The first time we start Minikube, we download the VM boot image.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\myuser> minikube start --driver=vmware
* minikube v1.9.1 on Microsoft Windows 10 Enterprise 10.0.17134 Build 17134
* Using the vmware driver based on user configuration
* Downloading VM boot image ...
  > minikube-v1.9.0.iso.sha256: 65 B / 65 B [-----] 100.00% ? p/s 0s
  > minikube-v1.9.0.iso: 174.93 MiB / 174.93 MiB [ 100.00% 1.36 MiB p/s 2m10s
* Starting control plane node m01 in cluster minikube
* Downloading Kubernetes v1.18.0 preload ...
  > preloaded-images-k8s-v2-v1.18.0-docker-overlay2-amd64.tar.lz4: 542.91 MiB
* Creating vmware VM (CPUs=2, Memory=4000MB, Disk=20000MB) ...
* Preparing Kubernetes v1.18.0 on Docker 19.03.8 ...
* Enabling addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube"
PS C:\Users\myuser>
```

Verify that the Kubernetes cluster is running

Check the Minikube's status.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\myuser> minikube status
m01
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
PS C:\Users\myuser>
```

Get the kubectl cluster info.

```
PS C:\Users\myuser> kubectl cluster-info
Kubernetes master is running at https://192.168.150.129:8443
KubeDNS is running at https://192.168.150.129:8443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy
```

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

```
PS C:\Users\myuser>
```

Minikube Start up and Clean Up

Here, we have an example Minikube start (with the VM boot image already installed), followed by stop and delete.

```
PS C:\Users\myuser> minikube start --driver=vmware
* minikube v1.9.1 on Microsoft Windows 10 Enterprise 10.0.17134 Build 17134
* Using the vmware driver based on user configuration
* Starting control plane node m01 in cluster minikube
* Creating vmware VM (CPUs=2, Memory=4000MB, Disk=20000MB) ...
* Preparing Kubernetes v1.18.0 on Docker 19.03.8 ...
* Enabling addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube"
PS C:\Users\myuser> minikube stop
* Stopping "minikube" in vmware ...
* Node "m01" stopped.
PS C:\Users\myuser> minikube delete
* Deleting "minikube" in vmware ...
* Removed all traces of the "minikube" cluster.
PS C:\Users\myuser>
```